

TECHNOLOGY & TOOLS REQUIREMENT PROPOSAL

AirSense

Urban Pollution Response System

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Project: CEP Group Project — AirSense

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"This neighbourhood is experiencing an abnormal pollution event. Here is the most likely cause, who or what should be investigated, what intervention is appropriate, and whether pollution actually decreased after the intervention."

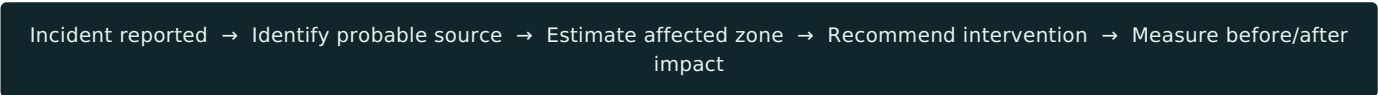
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1. Project Context & Reframed Objective

AirSense currently exists as a working frontend prototype (React + TypeScript, deployed as a single-page app) that displays AQI values, forecasts, and pollution source breakdowns using mock data and rule-based logic, with Google Gemini generating natural-language explanations. The project documentation already describes this data layer as "drop-in replaceable" with a live feed — the architecture was built anticipating this next step.

This proposal covers that next step, reframed around a sharper problem than a dashboard solves: **cities know pollution is bad, but they typically lack a hyperlocal, evidence-driven way to decide where to act, what is causing a specific incident, and whether an intervention actually worked.** A resident reporting dust or smoke today has no system that connects their complaint to a probable source, an estimated affected radius, a recommended action, and a measured outcome.



This is a measurable, documented gap. A Centre for Science and Environment analysis in the *State of India's Environment 2026* report found that only about 15% of India's population — roughly 200 million people — lives within 10 km of a continuous air-quality monitor, leaving over 1.2 billion people outside real-time coverage; the report separately flags hotspot-based planning, source assessment, and post-intervention evaluation as specific gaps in urban clean-air management. Pune is a strong pilot city for this problem: CPCB's local source-apportionment work and TERI's more recent Pune analysis both identify road dust, vehicle emissions, construction activity, and solid-fuel use as major PM contributors — exactly the categories AirSense's SourceAnalysis module needs to distinguish between.

Goal of this document: list the concrete tools, services, and data-access items required to move AirSense from a mock-data prototype to a real, usable urban pollution response system — one credible enough to eventually take to a municipal or state pollution-control body as a working solution.

2. Tools & Technology Required

Tools are grouped by function. Badges indicate whether an item is open-source/free, a paid item that needs budget approval, or an institutional access request the college can help secure.

A. AI Development & Research Assistant

Tool	Purpose	Why It's Needed for This Project
Claude Paid	Research synthesis, full-stack + ML code generation, documentation, and explainability review	Functions as the project's primary technical collaborator across the entire stack — see Section 3 for a full breakdown. The two supporting method-selection documents referenced throughout this proposal were themselves produced this way, demonstrating the workflow before it is funded.

B. Machine Learning & Data Science Stack

Tool	Purpose	Why It's Needed for This Project
Python 3.x Free	Runtime for all ML and data-processing code	The existing app is TypeScript-only; every recommended model family (Random Forest, XGBoost, Prophet, SHAP) lives in the Python ecosystem, so this is the base dependency for everything below.
scikit-learn Free	Random Forest Regressor/Classifier baselines	The single most-repeated best performer for tabular AQI prediction across the underlying literature review — the correct default for the 30-day AQI trend and the inversion-risk classifier.
XGBoost / LightGBM / CatBoost Free	Gradient-boosted trees for forecasting and source attribution	LightGBM is the recommended first model for the 24-hour multivariate forecast (cheapest to run, outperformed LSTM/CNN-LSTM in the reviewed study); XGBoost/CatBoost are the fallback for AQI regression and the base model for SourceAnalysis.
SHAP Free	Per-feature attribution on top of the tree models above	Turns SourceAnalysis from a static mock percentage into a real, explainable attribution of AQI drivers by category — the highest differentiation-per-effort feature identified for this project.
Facebook Prophet Free	Medium-term (7-day) seasonal forecasting	More interpretable and flexible for seasonal pollution patterns than ARIMA, which matters since each forecast day is paired with a recommendation string a resident will read.
pandas / NumPy Free	Data cleaning and transformation	Required to process whatever real feed (sensor, CPCB, satellite) replaces the current mock TypeScript arrays.
PyTorch Free <i>(later phase)</i>	LSTM / LSTM-Autoencoder models	Deliberately not needed at project start — deep learning needs data volume this project won't have until live feeds are running (see Phase 5, Section 4). Listed here so it isn't a surprise later.

C. Geospatial, Mapping & Environmental Data

Tool	Purpose	Why It's Needed for This Project
QGIS Free	Prototyping and validating affected-zone geometry	Needed to design and sanity-check the 1–2 km "affected zone" logic (buffers, wind-adjusted shapes) before it is coded into the product.
Google Maps Platform or Mapbox Paid (metered, free tier)	Geocoding + rendering affected-zone overlays	Converts a citizen's complaint (address or pin) into coordinates and renders the estimated impact zone on InteractiveMap — this is the visual centerpiece of the reframed product.
Meteorological data API (IMD / OpenWeatherMap) Free tier	Wind speed & direction feed	Wind data is a required input for both the inversion-risk classifier and any wind-informed affected-zone estimate — without it, "estimate affected area" cannot move past a static radius.
Sentinel-5P / NASA FIRMS satellite data Access request	Gridded aerosol/fire data for future spatial modeling	Explicitly the prerequisite the ML review identifies before InteractiveMap's sparse 20-city markers can become a real spatial-grid (CNN) task. Not needed immediately, but account access should be requested now since approval can take time.
CPCB SAMEER / PRANA data access Institutional request	Ground-truth complaint and source-category data	This is the actual evidence base the "verify the likely source" pipeline depends on. Formal access will almost certainly require an institutional request — this is the single item most likely to need the college's help, not just its budget.

D. Backend, Database & Infrastructure

Tool	Purpose	Why It's Needed for This Project
Node.js / Express.js Free	Existing optional backend layer	Already anticipated in the project's architecture; the frontend continues talking to this same API surface, so no frontend rework is required.
Python microservice (FastAPI/Flask) Free	Serves the scikit-learn/XGBoost models behind a REST endpoint	Tree-based models are trivial to serve this way and this keeps every frontend component completely unchanged — the exact split the project's own architecture note calls for.
PostgreSQL + PostGIS Free	Spatial-relational storage for incidents, readings, and zone polygons	The moment "estimate affected area" becomes a real geometry instead of a label, a spatial database is required — PostGIS is the standard, free choice for this.
Docker Free	Containerizes the Python ML service	Lets the ML sidecar be deployed and updated independently of the frontend, and makes the eventual handoff to another team (or a government IT department) far more straightforward.
Cloud hosting (student credits / small PaaS) Paid	Always-on deployment	Required to have an actual system "used by real users," as the project intends, rather than a build that only runs on a laptop.

E. Frontend & Existing Integrations (retained — no new spend)

Tool	Purpose	Why It's Needed for This Project
React + TypeScript + Vite Free	Existing frontend framework	Already built and intentionally designed so every ML swap happens behind these components — listed here only for completeness.
Google Gemini API Existing	Natural-language content for AwarenessHub	Already the right tool for quiz/article generation and out of scope for this ML work; no change requested.

F. Collaboration, Version Control & Documentation

Tool	Purpose	Why It's Needed for This Project
GitHub Free	Source control, issue tracking, project board	Needed once this moves from a solo mock-data build to a phased, fundable roadmap with reviewable history — important for any future team handoff.
Figma Free tier	UI polishing for map and recommendation views	The affected-zone map and intervention-recommendation screens are what a government stakeholder will actually look at — worth polishing deliberately before that meeting.

G. Research & Literature Access

Tool	Purpose	Why It's Needed for This Project
Science of the Total Environment (Elsevier) & related journals Institutional	Continued access to air-quality ML literature	The entire method-selection strategy in this proposal is derived from one 2025 review paper (Karmoude et al.); continued library access is needed as the project moves past that single source into specific implementation questions.
CPCB / TERI Pune source-apportionment reports Public	Local ground-truth for Pune's pollution sources	Already identified and directly usable, but formal citation/access rights may matter for any eventual government-facing submission.

H. Optional / Future — Field Validation Hardware

Tool	Purpose	Why It's Needed for This Project
Low-cost PM2.5/PM10 sensor kit (e.g. ESP32 + PMS-series sensor) Paid, one-time	A real low-cost sensor stream to calibrate against	Not required to start, but DataProvenance's "outlier rejection, humidity correction" work needs at least one live sensor feed to calibrate the anomaly-detection pipeline against — flagged as a Phase 4+ ask, not an immediate one.

3. The Role of Claude in This Project

Claude is listed as its own line item in Section 2 because, for a solo-founder technical build like this one, it functions less like a single "tool" and more like a second engineer — one who can move across research, code, geospatial logic, and documentation in the same session. The two supporting documents that inform this proposal's technical recommendations were themselves produced using Claude, which is offered here as evidence of the workflow rather than a claim about it.

3.1 Research & Literature Synthesis

Academic reviews like the 2025 *Science of the Total Environment* paper underpinning this project's method choices run to dozens of studies and hundreds of data points. Claude condensed that single review into a feature-by-feature mapping against AirSense's actual modules — which is the difference between "read a paper" and "know what to build in CityDetail versus SourceAnalysis." The same synthesis work applies going forward as the project reads CPCB/TERI source-apportionment reports, CSE monitoring-gap assessments, and further ML literature.

3.2 Full-Stack + ML Development, via Claude Code

A Pro or Max subscription includes Claude Code, an agentic coding tool that works directly in the project's terminal or IDE. For this build specifically, that means Claude can scaffold the Python FastAPI microservice, wire SHAP output into the SourceAnalysis component, write the PostGIS queries behind the affected-zone geometry, and debug across the existing TypeScript frontend and the new Python backend — in the same session, without the developer manually shuttling context between two separate tools.

3.3 Model Interpretability & Explainability Review

The underlying literature review is explicit that health- and policy-facing predictions need to stay explainable, not just accurate — this is precisely why Random Forest/SHAP was chosen over a black-box deep model for the Simulator and SourceAnalysis modules. Claude can review SHAP or feature-importance output and turn it into the kind of plain-language explanation a resident or a municipal officer needs — not just a technically correct one.

3.4 Documentation & Government-Facing Communication

As this moves from a college project toward a pitch a government pollution-control body might actually act on, it will need clean technical documentation, formal data-access requests (e.g., to CPCB/MPCB for SAMEER or PRANA access), and presentation material a non-technical official can follow. This proposal document is itself an example of that output.

3.5 Recommended Plan: Pro vs. Max

Claude Pro is priced at \$20/month (US), with local-currency pricing available in supported regions, and includes both Claude Code and Cowork access alongside materially higher usage limits than the free tier. Claude Max provides five or twenty times Pro's usage ceiling, at the corresponding higher price, along with priority access to new models and features — the same Claude Code and Cowork access, just with more headroom for long, daily sessions.

Plan	What It Covers	Fit for This Project
Claude Pro	Claude Code, Cowork, ~5x free-tier usage per session, priority access during high-traffic periods	Sufficient for Phases 1–3 (Section 4): forecasting models, SHAP integration, and the geospatial affected-zone logic. This is the item this proposal requests funding for now.
Claude Max	5x or 20x Pro's usage ceiling, priority access to new models, same Claude Code + Cowork access	Worth upgrading to once the project enters Phase 4–5 (live data integration, LSTM work, longer daily agentic-coding sessions) or once more than one contributor is actively building — a Phase-2 budget line, not an immediate one.

Pricing and plan details per Anthropic's official support documentation (support.claude.com), current as of August 2026; confirm current figures at signup, as plans and pricing are subject to change.

4. Phased Build Plan

Tools are not all needed on day one. This sequencing follows the same "build in order" logic established in the project's ML method-selection review, extended to include the geospatial incident-response layer this proposal adds.

Phase 1 — Core Prediction Backbone

Tools: Python, scikit-learn, XGBoost/LightGBM, FastAPI, Claude Pro.

Goal: Replace CityDetail and Dashboard's mock forecasts and the 30-day AQI trend with real Random Forest/XGBoost models — tabular, cheap, no GPU required, and the most-validated method family in the underlying review. One model quietly upgrades three modules (CityDetail, Dashboard, and AlertsManager via its output) at once.

Phase 2 — Source Attribution Layer

Tools: SHAP, on top of the Phase 1 model.

Goal: Turn SourceAnalysis into a real per-category attribution (road dust / vehicle / construction / solid-fuel) instead of a static mock percentage — no new model family, just an interpretability layer on what Phase 1 already built.

Phase 3 — Geospatial "Affected Zone" & Intervention Loop

Tools: PostGIS, QGIS, Google Maps/Mapbox, meteorological API.

Goal: Convert a reported incident into coordinates, generate a wind-informed affected-zone polygon, and log a before/after AQI comparison once an intervention is applied. This is the phase where the reframed problem statement — "PM10 reduced by 28% after intervention" — becomes an actual, storable outcome rather than an example.

Phase 4 — Anomaly Detection & Community Corroboration

Tools: Statistical outlier removal first; LSTM-Autoencoder + One-Class SVM later.

Goal: DataProvenance's calibration work and CommunityFeed's discrepancy flagging share one anomaly-detection pipeline — one build unlocks two features, matching the project's own roadmap item for automated report validation.

Phase 5 — Live Data & Deep Learning

Tools: Formal CPCB/SAMEER/PRANA access, PyTorch/LSTM, Claude Max, cloud hosting at scale.

Goal: Swap real feeds in behind the existing components. Only once live time-series volume exists does LSTM/hybrid modeling earn its computational cost — building it earlier would just overfit to a small, synthetic city set.

Phase 6 — Satellite / Spatial Layer

Tools: Sentinel-5P / NASA FIRMS access, CNN.

Goal: InteractiveMap moves from 20 discrete markers to real gridded spatial inference — held off deliberately until this data exists, since 20 sparse points is a visualization problem, not yet an ML one.

5. Budget & Licensing Summary

Most of the technical stack is open-source and carries no licensing cost. The items below are the actual budget and support ask.

Item	Type	Indicative Cost	Notes
Claude Pro subscription	Paid	~\$20/month (local-currency pricing may apply)	Core AI development & research tool; scale to Max in a later phase
Cloud hosting	Paid	Low, or covered by student/education credits	Small VM or PaaS tier is sufficient through Phase 4
Google Maps Platform / Mapbox	Paid (metered)	Likely within free tier at pilot scale	Geocoding + affected-zone rendering
Python/ML/backend stack	Open-source	Free	scikit-learn, XGBoost, SHAP, Prophet, FastAPI, PostgreSQL/PostGIS, Docker
GitHub, Figma	Free tier	Free	Version control and UI design
Journal / library access	Institutional	Via college library subscription	Continued access to air-quality ML literature
CPCB SAMEER / PRANA data access	Institutional request	No direct fee; needs formal approval	Core data dependency — the college's support in requesting this matters more than budget
Low-cost sensor kit (Phase 4+)	Paid, one-time	Modest, per unit	Optional; deferred until the software pipeline is stable

6. Request to the College

This proposal requests:

1. **Funding for one Claude Pro subscription** to begin Phase 1–3 development, with Claude Max as a follow-on request once the project reaches Phase 4–5 or expands beyond one contributor.
2. **A small cloud hosting budget** (or guidance toward available student/education credits) so the system is deployed and reachable, not just running locally.
3. **Institutional support in requesting formal data access** from CPCB (SAMEER/PRANA) and, where relevant, MPCB — this is the item most likely to require the college's standing rather than its budget.
4. **Continued library access** to the air-quality ML literature this project's method choices are built on.

All core ML, backend, and geospatial tooling is open-source and requires no license fee — the items above represent the complete budget ask needed to take AirSense from its current mock-data prototype to a real, evidence-driven urban pollution response system.

Supporting technical documents: *Best-Performing ML Methods by Data Type and Task in Air Quality Prediction* and *ML Method Recommendations for AirSense — Feature-by-Feature Mapping*, both available on request, provide the full method-selection reasoning summarized in Sections 2 and 4 of this proposal.